

# Deep Generative Models

## Lecture 8

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# Recap of Previous Lecture

## Forward Gaussian Diffusion Process

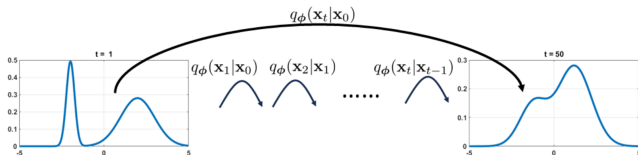
Let  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ,  $\beta_t \ll 1$ ,  $\alpha_t = 1 - \beta_t$  and  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ .

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, \mathbf{I});$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

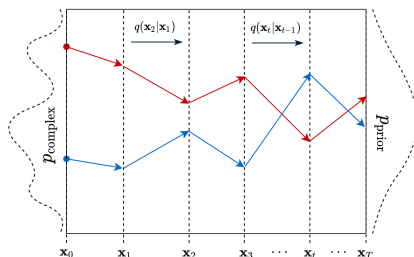
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}).$$



# Recap of Previous Lecture

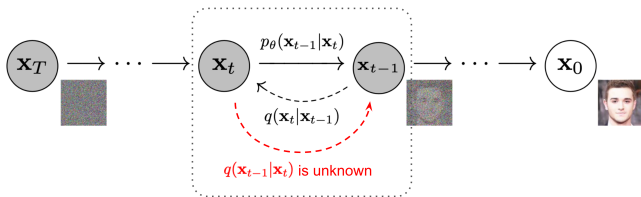
**Diffusion** describes the process where particles migrate from regions of high density to regions of low density.



1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ;
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$ , with  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ ,  $t \geq 1$ ;
3.  $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ , for  $T \gg 1$ .

If we can invert this process, we would have a way to sample  $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$  using noise samples, i.e.  $p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ . Hence, our objective becomes to reverse this process.

# Recap of Previous Lecture



## Reverse Process (Ancestral Sampling)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

The Feller theorem guarantees this approximation is valid.

## Forward Process

1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ;
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$ ;
3.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ .

## Reverse Process

1.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ ;
2.  $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$ ;
3.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ .



## Recap of Previous Lecture

**Forward Process:** Converts an arbitrary distribution  $p_{\text{data}}(\mathbf{x})$  into the standard normal  $\mathcal{N}(0, \mathbf{I})$  by incrementally adding noise.

$$\begin{aligned}q(\mathbf{x}_t | \mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}); \\q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}).\end{aligned}$$

**Reverse Process:** This intractable distribution can be well-approximated by a Gaussian (with unknown parameters) when  $\beta_t$  is sufficiently small.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

**Conditioned Reverse Process:** This Gaussian, with known parameters, describes how to denoise a noisy image  $\mathbf{x}_t$  when the final clean image  $\mathbf{x}_0$  is known.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I})$$

## Recap of Previous Lecture

- ▶  $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$  represents the latent variables.
- ▶ Variational posterior distribution:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

- ▶ Generative model and prior:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = p_{\theta}(\mathbf{x}_0|\mathbf{x}_1); \quad p_{\theta}(\mathbf{z}) = \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T)$$

## ELBO

$$\log p_{\theta}(\mathbf{x}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\phi, \theta}(\mathbf{x}) \rightarrow \max_{q, \theta}$$

$$\begin{aligned} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t} \end{aligned}$$

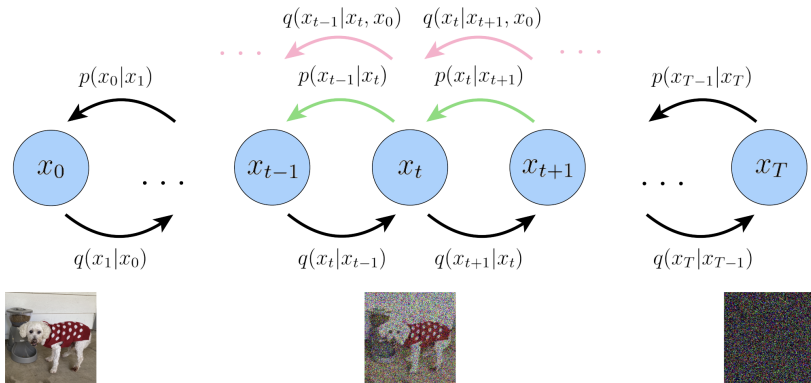
# Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
  - Classifier Guidance
  - Classifier-Free Guidance

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# ELBO for Gaussian Diffusion Model



$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}),$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

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Let's assume that

$$\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t) = \tilde{\beta}_t \mathbf{I} \quad \Rightarrow \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}).$$

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Theoretically, the optimal  $\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t)$  lies in  $[\tilde{\beta}_t, \beta_t]$ :

- ▶  $\beta_t$  is optimal for  $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$ ;
- ▶  $\tilde{\beta}_t$  is optimal for  $\mathbf{x}_0 \sim \delta(\mathbf{x}_0 - \mathbf{x}^*)$ .



# ELBO for Gaussian Diffusion Model

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# ELBO for Gaussian Diffusion Model

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute noisy image  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$ .
3. Compute the ELBO:

$$\begin{aligned} \mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1 | \mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ & - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta, t}(\mathbf{x}_t)\|^2 \right]}_{\mathcal{L}_t}. \end{aligned}$$

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## Sampling

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise  $\mathbf{x}_{t-1} = \mu_{\theta, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

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# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

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- ▶ There is a linear relationship between  $\boldsymbol{\epsilon}$ ,  $\mathbf{x}_t$ , and  $\mathbf{x}_0$ .
- ▶ Let's try to rewrite this mean using only  $\mathbf{x}_t$  and  $\boldsymbol{\epsilon}$ .

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$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \boldsymbol{\epsilon}) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left( \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}} \right) \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \cdot \boldsymbol{\epsilon} \end{aligned}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

## Reparametrization

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t)\end{aligned}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

## Reparametrization

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon$$

$$\mu_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t)$$

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

# Reparametrization of DDPM

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## Reparametrization

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon$$

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# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

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At every step of the reverse process, we attempt to predict the noise  $\epsilon$  that was used in the forward diffusion process!



# Reparametrization of DDPM

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

# Reparametrization of DDPM

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$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) - \\ - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}$$

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2 \right]$$

Let's drop the scaling coefficient.

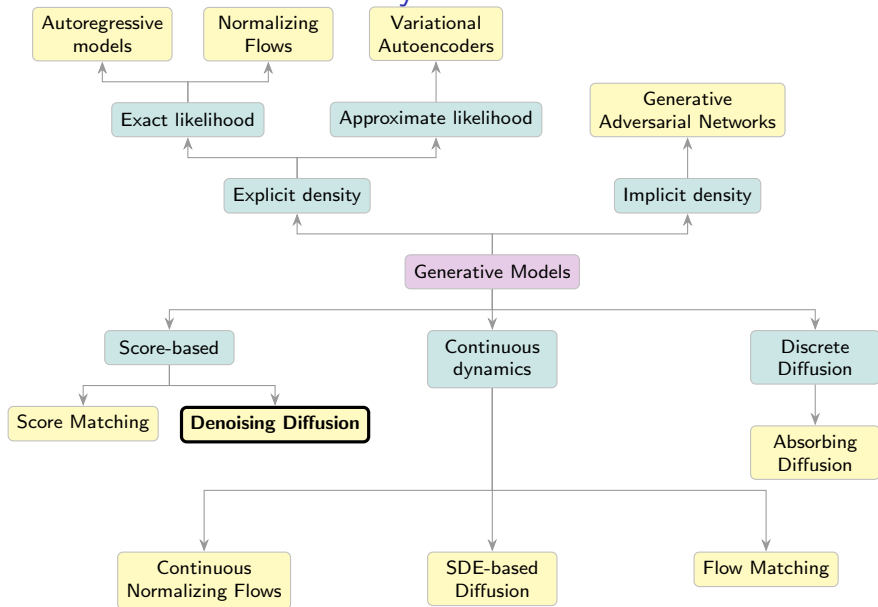
## Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{2, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon) \right\|^2$$

# Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
  - Classifier Guidance
  - Classifier-Free Guidance

# Generative Models Taxonomy



# Denoising Diffusion Probabilistic Model (DDPM)

## DDPM is a VAE Model

- ▶ The encoder is a fixed Gaussian Markov chain  $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$ .
- ▶ The latent variable is hierarchical (at each step, its dimension equals the input's).
- ▶ The decoder is a simple Gaussian model  $p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)$ .
- ▶ The prior distribution is given by a parametric Gaussian Markov chain  $p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$ .

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## Forward Process

1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ;
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon$ ;
3.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ .

# Denoising Diffusion Probabilistic Model (DDPM)

## DDPM is a VAE Model

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3.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ .

## Reverse Process

1.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ ;
2.  $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$ ;
3.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ .



# Denoising Diffusion Probabilistic Model (DDPM)

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ .
2. Sample  $t \sim U\{1, T\}$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
3. Compute noisy image  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$ .
4. Compute loss  $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$ .

# Denoising Diffusion Probabilistic Model (DDPM)

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4. Compute loss  $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$ .

## Sampling (Ancestral)

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute the mean of  $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$ :

$$\mu_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t).$$

3. Denoise  $\mathbf{x}_{t-1} = \mu_{\theta,t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

# Denoising Diffusion as a Score-Based Generative Model

## DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{1,t} \cdot \|\epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon\|_2^2 \right]$$

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# Denoising Diffusion as a Score-Based Generative Model

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# Denoising Diffusion as a Score-Based Generative Model

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We can reparameterize the model as:

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t) = -\frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t).$$

# Denoising Diffusion as a Score-Based Generative Model

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$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{1,t} \cdot \|\epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon\|_2^2 \right] \\ &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{2,t} \cdot \left\| \frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} - \frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \right\|_2^2 \right] \\ q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}) \\ \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) &= -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}}.\end{aligned}$$

We can reparameterize the model as:

$$\begin{aligned}\mathbf{s}_{\theta,t}(\mathbf{x}_t) &= -\frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t). \\ \mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ C_{2,t} \cdot \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]\end{aligned}$$



# DDPM vs NCSN: Objectives

## DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

# DDPM vs NCSN: Objectives

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$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

# DDPM vs NCSN: Objectives

## DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$$

In practice, **this coefficient** is often omitted.

## NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta,\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

# DDPM vs NCSN: Objectives

## DDPM Objective

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In practice, **this coefficient** is often omitted.

## NCSN Objective

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$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

**Maximizing the ELBO leads to the same objective as denoising score matching!**

# DDPM vs NCSN: Sampling

## DDPM Sampling (Ancestral Sampling)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mu_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

# DDPM vs NCSN: Sampling

## DDPM Sampling (Ancestral Sampling)

$$\begin{aligned}\mathbf{x}_T &\sim \mathcal{N}(0, \mathbf{I}) \\ \mathbf{x}_{t-1} &= \mu_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon\end{aligned}$$

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# DDPM vs NCSN: Sampling

## DDPM Sampling (Ancestral Sampling)

$$\begin{aligned}\mathbf{x}_T &\sim \mathcal{N}(0, \mathbf{I}) \\ \mathbf{x}_{t-1} &= \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon} \\ &= \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}\end{aligned}$$

## NCSN Sampling (Annealed Langevin Dynamics)

- ▶ Sample  $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$ .
- ▶ Perform  $L$  steps of Langevin dynamics:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \cdot \mathbf{s}_{\theta,\sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \cdot \boldsymbol{\epsilon}_t^l.$$

- ▶ Set  $\mathbf{x}_{t-1}^0 = \mathbf{x}_t^L$  and move to the next  $\sigma_t$ .



# DDPM vs NCSN: Summary

## Summary

- ▶ Different Markov chains:
  - ▶ DDPM:  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
  - ▶ NCSN:  $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
  - ▶ One can generalize to  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$

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*Kingma D. et al. Variational Diffusion Models, 2021*

*Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# DDPM vs NCSN: Summary

## Summary

- ▶ Different Markov chains:
  - ▶ DDPM:  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
  - ▶ NCSN:  $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
  - ▶ One can generalize to  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- ▶ The objectives coincide: ELBO  $\equiv$  score-matching.

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*Kingma D. et al. Variational Diffusion Models, 2021*

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# DDPM vs NCSN: Summary

## Summary

- ▶ Different Markov chains:
  - ▶ DDPM:  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
  - ▶ NCSN:  $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
  - ▶ One can generalize to  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- ▶ The objectives coincide: ELBO  $\equiv$  score-matching.
- ▶ The sampling procedures differ:
  - ▶ Ancestral sampling in DDPM;
  - ▶ Annealed Langevin dynamics for NCSN;
  - ▶ Hybrid approaches that combine both updates are possible.

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*Kingma D. et al. Variational Diffusion Models, 2021*

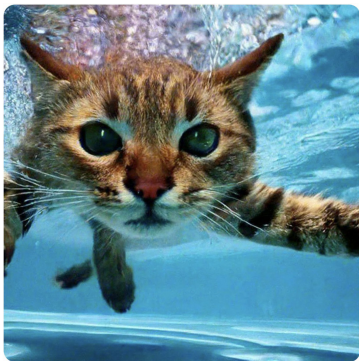
*Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020*

# Outline

1. Diffusion ELBO Derivation (continued)
2. Gaussian Diffusion Reparametrization
3. Denoising Diffusion Probabilistic Model (DDPM)
4. Model Guidance
  - Classifier Guidance
  - Classifier-Free Guidance

## Guidance

- ▶ Up to now, we have focused on **unconditional** generative models  $p_{\theta}(\mathbf{x})$ .
- ▶ In practice, most generative models are **conditional** (in the diffusion era it is called guided):  $p_{\theta}(\mathbf{x}|\mathbf{y})$ .
- ▶ Here,  $\mathbf{y}$  might denote a class label or **text** (as in text-to-image tasks).



Кот ныряет в бассейн, как ребенок на обложке альбома Nevermind, реалистично



рука человека с пятью пальцами, ни четырьмя, ни шестью, а с 5 (пять) пальцами

# Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution  $p_{\text{data}}(\mathbf{x}|\mathbf{y})$ ).

- ▶  $\mathbf{y} = \emptyset$ ,  $\mathbf{x} = \text{image}$   $\Rightarrow$  unconditional image model
- ▶  $\mathbf{y} = \text{class label}$ ,  $\mathbf{x} = \text{image}$   $\Rightarrow$  class-conditional image model
- ▶  $\mathbf{y} = \text{text prompt}$ ,  $\mathbf{x} = \text{image}$   $\Rightarrow$  text-to-image model
- ▶  $\mathbf{y} = \text{image}$ ,  $\mathbf{x} = \text{image}$   $\Rightarrow$  image-to-image model
- ▶  $\mathbf{y} = \text{image}$ ,  $\mathbf{x} = \text{text}$   $\Rightarrow$  image-to-text (image captioning) model
- ▶  $\mathbf{y} = \text{English text}$ ,  $\mathbf{x} = \text{Russian text}$   $\Rightarrow$  sequence-to-sequence model (machine translation)
- ▶  $\mathbf{y} = \text{sound}$ ,  $\mathbf{x} = \text{text}$   $\Rightarrow$  speech-to-text (automatic speech recognition) model
- ▶  $\mathbf{y} = \text{text}$ ,  $\mathbf{x} = \text{sound}$   $\Rightarrow$  text-to-speech model

# Label Guidance

**Label:** Ostrich (10th ImageNet class)



**VQ-VAE (Proposed)**

**BigGAN deep**

# Text Guidance

**Prompt:** a stained glass window of a panda eating bamboo

Left:  $\gamma = 1$ , Right:  $\gamma = 3$ .



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Nichol A., et al. *GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models*, 2022



# Guidance in Generative Models

How to make a guided model?

Instead of sampling from  $p_{\theta}(\mathbf{x})$ , we sample from  $p_{\theta}(\mathbf{x}|\mathbf{y})$ .

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- ▶  $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$  for DDPMs.

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- ▶  $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$  for DDPMs.

## Challenge

- ▶ Empirically, images sampled with this procedure do not fit well enough to the desired label  $\mathbf{y}$ .
- ▶ Being able to control the strength of guidance is especially valuable.

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# Classifier Guidance

## DDPM Sampling

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Generate the denoised image (unconditional generation):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

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## Guided Generation

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What is the link between  $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)$  and  $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t|\mathbf{y})$ ?

# Classifier Guidance: Guided Score Function

## Guided Generation

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# Classifier Guidance: Guidance Scale

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## Guidance Scale

It is a natural idea to scale up the contribution of the guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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- ▶ The **guidance scale**  $\gamma$  adjusts the strength of classifier guidance.
- ▶  $\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y})$  is not the true guided score function  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Classifier Guidance: Distribution Sharpening

## Scaled Guided Score Function

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**Note:** Increasing  $\gamma$  sharpens  $p(\mathbf{y}|\mathbf{x}_t)$ , increasing contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

# Classifier Guidance: Overview

## Training

1. Train the DDPM as before.
2. Train an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  on noisy data (time-dependent).

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## Sampling (Guided)

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

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# Classifier-Free Guidance

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Bayes theorem

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$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log \left( \frac{p_\theta(\mathbf{x}_t|\mathbf{y})p(\mathbf{y})}{p_\theta(\mathbf{x}_t)} \right) \\ &= \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) \end{aligned}$$

## Scaled Guided Score Function

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# Classifier-Free Guidance

- ▶ The previous approach relies on training an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  for noisy images.
- ▶ We now introduce a method to sidestep this requirement.

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# Classifier-Free Guidance: Formulation

## Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$$

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## Naive training approach

- ▶ Train an unguided score function model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$ .
- ▶ Train a guided score function model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .



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- ▶ Use their convex combination at inference.

## Guided Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

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How to avoid training two separate score function models?

# Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

## CFG

1. Introduce the “absence of conditioning” label  $\mathbf{y} = \emptyset$ .
2. Identify the unguided score function  $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ .
3. Train a single model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$  on **supervised** data, dropping the label  $\mathbf{y}$  with some fixed probability (simulating  $\mathbf{y} = \emptyset$ ).
4. At inference, evaluate the model twice to obtain  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$  and  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

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## Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

# Summary

- ▶ At each step, DDPM predicts the noise that was injected in the forward process.
- ▶ DDPMs are quite slow, since the model must be applied  $T$  times for sampling.
- ▶ DDPM and NCSN are intimately connected at the objective level.
- ▶ Guidance technique allows to make controllable generation via conditioning on labels or text prompts.
- ▶ Classifier guidance turns an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- ▶ Classifier-free guidance removes the need for such a classifier, yielding a practical recipe now widely used.